THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 4: **Expressions using Letter-Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A taxi charges ₹60 fixed plus ₹10 per km. For $k = 8$ km the fare is (₹): 1

(a) 141 **(b)** 140 **(c)** 139 **(d)** 142

2. The perimeter of a square of side s is $4s$. For $s = 15$ cm it is (cm): 1

(a) 60 **(b)** 61 **(c)** 62 **(d)** 59

3. The nth odd number is: 1

(a) $n - 1$ **(b)** $2n$ **(c)** $2n - 1$ **(d)** $2n + 2$

4. If $m = 3$, the value of $m + 7m$ is: 1

(a) 25 **(b)** 23 **(c)** 26 **(d)** 24

5. Subtract $11x$ from $20x$: the result is: 1

(a) $-9x$ **(b)** $31x$ **(c)** 9 **(d)** $9x$

6. The value of $5(x + 3)$ for $x = 2$ is: 1

(a) 27 **(b)** 25 **(c)** 26 **(d)** 24

7. 'Product of p and q, increased by 5' is: 1

(a) $pq + 5$ **(b)** $5pq$ **(c)** $p + q + 5$ **(d)** $p(q + 5)$

8. A taxi charges ₹30 fixed plus ₹18 per km. For $k = 3$ km the fare is (₹): 1

(a) 84 **(b)** 86 **(c)** 85 **(d)** 83

9. The perimeter of a square of side s is $4s$. For $s = 20$ cm it is (cm): 1

(a) 79 **(b)** 81 **(c)** 82 **(d)** 80

10. 'Half of the sum of a and b' is: 1

(a) $a/2b$ **(b)** $2(a + b)$ **(c)** $(a + b)/2$ **(d)** $a + b/2$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the value of $8p - 2q$ for $p = 5$, $q = 3$. 2

12. Find the number of matchsticks needed for 12 squares in a row if each square needs 4 sticks but neighbours share one: use $3n + 1$. 2

13. Simplify: $10x + 9 - 4x + 1$. 2

14. Find the value of $6p - 5q$ for $p = 7$, $q = 6$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In a pattern, the number of dots in figure n is $3n + 2$. Find the dots in figure 5, and which figure has 47 dots. 3

16. Add $2x + 10y$ and $8x - 7y$. 3

17. The age of a father is 2 times his son's age s plus 6 years. Write the expression, and find the father's age when $s = 16$. 3

OR

In a pattern, the number of dots in figure n is $3n + 2$. Find the dots in figure 12, and which figure has 20 dots. 3

18. Add $6x + 4y$ and $5x - 3y$. 3

OR

The age of a father is 3 times his son's age s plus 1 years. Write the expression, and find the father's age when $s = 11$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A school buys n notebooks at ₹18 each and 2 pens at ₹10 each. Write the total cost, and find it when $n = 9$. If ₹250 is paid, find the change. 5

OR

A hall has 9 rows of r chairs and 3 extra chairs at the back. Write the total number of chairs. Find it for $r = 11$, and the value of r if the total is 120. 5

20. A school buys n notebooks at ₹14 each and 3 pens at ₹7 each. Write the total cost, and find it when $n = 15$. If ₹300 is paid, find the change. 5

OR

A hall has 7 rows of r chairs and 9 extra chairs at the back. Write the total number of chairs. Find it for $r = 14$, and the value of r if the total is 135. 5

END OF QUESTION PAPER — Total: 40 Marks

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